PART I: FOUNDATIONS

Topic 4: Time Units, Frequency, and Aggregation

The Core Concept

- **Frequency:** How often we record data — hourly, daily, weekly, monthly, quarterly, or yearly.
- **Aggregation:** Grouping smaller time units into larger ones by summing or averaging (e.g., days → months).

Why It Matters

The frequency we choose can change our conclusions entirely:

- **Too granular (daily/hourly):** The data looks like random static. Every quiet night makes it look like crime has disappeared.
- **Too coarse (yearly):** A three-month spike that started and ended within the year becomes invisible.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Month	Precinct North (UoF)	Precinct South (UoF)	North Arrests
Jan	12	25	400
Feb	10	22	380
Mar	14	28	450
Apr	15	24	460
May	18	26	550
Jun	22	24	600
Jul	25	27	700
Aug	**(Missing)**	23	680
Sep	19	22	500
Oct	16	**85 ★ Outlier**	420
Nov	14	24	390
Dec	11	26	350

For Precinct North:

- **Yearly view:** "185 incidents total." The precinct seems consistently busy — no story.
- **Daily view:** "0, 1, 0, 0, 2, 0..." The data looks completely random and uninterpretable.
- **Monthly view:** The "mountain" pattern (winter low → summer peak → winter low) becomes clearly visible.

💡 **The Insight:** Monthly aggregation is often the "sweet spot" for public safety data — frequent enough to detect policy effects, smooth enough to filter out day-to-day noise.

Key Takeaway: Always match your time frequency to the question you are asking. Different frequencies can tell very different stories from the same data.